



Calculator App + GSAP



We took the original Project 4 calculator and enhanced it by integrating GSAP



Libraries we added:



GSAP 3.15



```
<script src="https://cdn.jsdelivr.net/npm/gsap@3.15/dist/gsap.min.js"></script>
```



```
<script src="https://cdn.jsdelivr.net/npm/gsap@3.15/dist/SplitText.min.js"></script>
```

1- SplitText Title Animation

<https://gsap.com/docs/v3/Plugins/SplitText/>

SplitText is a GSAP plugin that automatically splits any text element into individual characters, words, or lines so we can animate each one separately.

We used it to animate the title — each character rises up and fades in one by one with a small delay between them.

```
// https://gsap.com/docs/v3/Plugins/SplitText/
// we have to register SplitText so we can split the title into characters
gsap.registerPlugin(SplitText);

// we split the title
let split = SplitText.create(".split", { type: "words, chars" });

// we animate each character in with a stagger
gsap.from(split.chars, {
  duration: 1,
  y: 100,
  autoAlpha: 0,
  stagger: 0.05
});
```

Our Best Calc

Our Best Calculator App in the world

Our Best Calculator App in the world

2- Magnetic Button

<https://demos.gsap.com/demo/magnetic-button-overwrite>

The Start button on the start page follows the mouse when you hover over it, then snaps back with an elastic bounce when you leave. We use `gsap.utils.mapRange()` to calculate how far the button should move based on the mouse position inside the zone, then `gsap.to()` to animate it there smoothly.

```
// wiggle loop – button rocks back and forth
gsap.to(btn, {
  rotation: 12,
  duration: 0.15,
  repeat: -1,
  yoyo: true,
  // got sine.inOut from//
  // https://gsap.com/docs/v3/Eases/
  ease: "sine.inOut"
});

// magnetic pull – button follows the mouse inside the zone
zone.addEventListener("mousemove", (e) => {
  const rect = zone.getBoundingClientRect();
  const x = gsap.utils.mapRange(rect.left, rect.right, -rect.width / 2, rect.width / 2, e.clientX - rect.left);
  const y = gsap.utils.mapRange(rect.top, rect.bottom, -rect.height / 2, rect.height / 2, e.clientY - rect.top);

  gsap.to(btn, {
    x: x * strength,
    y: y * strength,
    duration: 0.4,
    ease: "power2.out",
    overwrite: "auto"
  });
});

// snap back when mouse leaves
zone.addEventListener("mouseleave", () => {
  gsap.to(btn, {
    x: 0,
    y: 0,
    duration: 0.7,
    ease: "elastic.out(1, 0.4)",
    overwrite: "auto"
  });
});
```

Start

Start

3- Button Spin on Click

[https://gsap.com/docs/v3/GSAP/gsap.to\(\)/](https://gsap.com/docs/v3/GSAP/gsap.to()/)

Every time a calculator button is clicked it does a full 360 degree spin. We use `gsap.to()` which animates an element TO the values you give it. The `+=360` means add 360 degrees on every click so it spins every time, not just the first.


```
// https://gsap.com/docs/v3/GSAP/gsap.to\(\)/  
// we use gsap.to() to spin the button on every click  
  
gsap.to(button, { rotation: "+=360", duration: 0.4 });  
  
if (value === "C") handleClear();  
else if (value === "backspace") handleBackspace();  
else if (value === "=") handleEquals();  
else if (operatorKeys.includes(value)) handleOperator(value);  
else handleNumber(value);  
  
});  
});
```

8			
C		←	÷
7	8	9	×
4	5	6	−
1	2	3	+
0			=

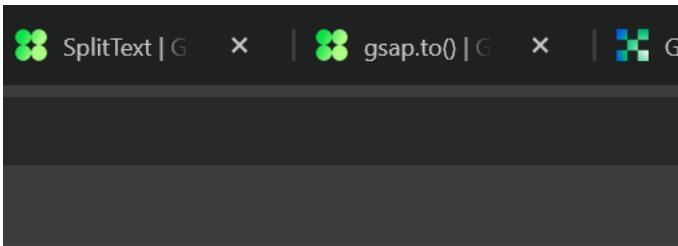
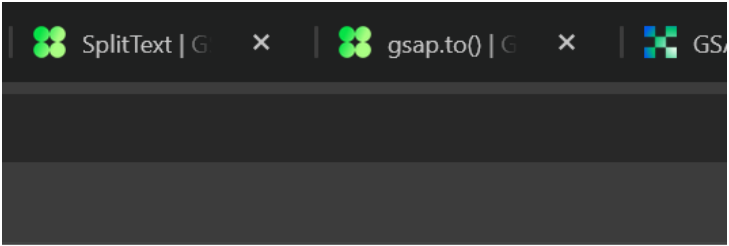
8			
C		←	÷
7	8	9	×
4	5	6	−
1	2	3	+
0			=

4-Page Load Animation

[https://gsap.com/docs/v3/GSAP/gsap.from\(\)/](https://gsap.com/docs/v3/GSAP/gsap.from()/)

When the calculator page loads, the whole calculator slides up from below and fades in. We use `gsap.from()` which animates FROM the values you give it back to the element's normal position. So it starts invisible and 100px below, then animates up over 2 seconds.

```
// https://gsap.com/docs/v3/GSAP/gsap.from\(\)  
// we use gsap.from() to slide the calculator up when we load the page  
  
gsap.from(".wrapper", { opacity: 0, y: 100, duration: 2.0,});
```



0			
C		←	÷
7	8	9	×
4	5	6	-
1	2	3	+
0			=

0			
C		←	÷
7	8	9	×
4	5	6	-
1	2	3	+
0			=

Live Demo

<https://ahmad-shawkhah.github.io/Calculator-with-animation/>